f_par_pro_WiSe 2024/25;

2. Aufgabe: Matrix-Matrix Multiplikation;

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Verwendete Hardware:

- HP Victus TG02-2301ng
- NVIDIA® GeForce RTX™ 4060 (8 GB GDDR6 dediziert)

Verwendete Programmierung-Umgebung: Microsoft Visual Studio auf Windows 11 Home

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Performance-Vergleich CudaMallocHost vs. Malloc ohne Shared-Memory

Werte: (CudaMallocHost-Malloc)/CudaMallocHost

1024	Min	Max	Avg	Median	StdDV
time*	6,75%	11,99%	5,91%	4,98%	25,40%
GFLOPS	-13,62%	-7,24%	-6,22%	-5,24%	9,51%
Memcpy AB	0,63%	12,50%	-1,75%	-1,07%	16,40%
Memcpy C	60,35%	68,94%	59,85%	58,94%	82,66%
Kernel	-0,10%	-2,26%	-0,62%	-0,80%	-17,72%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	-356,94%	-22,08%	100,00%	-139,76%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

4096	Min	Max	Avg	Median	StdDV
time*	1,61%	1,37%	1,50%	1,51%	-10,39%
GFLOPS	-1,39%	-1,63%	-1,53%	-1,54%	-13,54%
Memcpy AB	-2,88%	2,55%	-0,46%	-1,90%	27,48%
Memcpy C	58,73%	63,44%	60,45%	59,77%	92,38%
Kernel	0,05%	-0,08%	-0,03%	0,05%	-26,16%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	10,59%	-4,80%	-3,51%	-3,78%	-4,72%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

2048	Min	Max	Avg	Median	StdDV
time*	3,51%	1,85%	3,47%	3,64%	-38,18%
GFLOPS	-1,88%	-3,64%	-3,61%	-3,77%	-46,40%
Memcpy AB	0,35%	-8,02%	0,60%	3,38%	-39,68%
Memcpy C	59,60%	69,44%	59,55%	58,64%	78,69%
Kernel	-0,08%	0,34%	-0,08%	-0,03%	21,96%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	-38,87%	-19,55%	-15,36%	-6,99%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

8192	Min	Max	Avg	Median	StdDV
time*	0,92%	1,12%	1,00%	0,92%	25,69%
GFLOPS	-1,13%	-0,93%	-1,01%	-0,93%	24,21%
Memcpy AB	0,47%	2,92%	1,00%	2,21%	-0,08%
Memcpy C	58,11%	61,67%	59,15%	58,80%	95,57%
Kernel	0,15%	0,39%	0,24%	0,15%	24,42%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	-19,18%	8,11%	-2,79%	1,89%	0,04%
Free	0,00%	-36,75%	-36,75%	0,00%	-36,75%

Bei der Matrix-Größe von 1024 ist ein Leistungsgewinn von ca. 6 % (Zeit) bwz.ca. 6 % (GFLOPS) anfangs beachtlich. Mit zunehmender Matrix-Größe sinkt dieser deutlich. Bereits bei 2048 beträgt er nur noch knapp 3,5 % (Zeit und GFLOPS), bei 4096 nur noch 1,5 % (Zeit und GFLOPS), bei 8192 fällt er auf1 % (Zeit und GFLOPS). Ein wesentlicher Faktor ist die deutlich verkürzte Zeit Memcpy-Zeit für C, welche allgemein bei ca. 60 % liegt. Die Differenzen in den übrigen Größen (Kernel, Sync, Free, Malloc, Memcpy AB) dürften zufällig sein.

Performance-Vergleich CudaMallocHost vs. Malloc mit Shared-Memory

Werte: (CudaMallocHost-Malloc)/CudaMallocHost

1024	Min	Max	Avg	Median	StdDV
time*	13,97%	32,02%	16,50%	15,01%	84,10%
GFLOPS	-47,10%	-16,24%	-19,36%	-17,67%	73,91%
Memcpy AB	-0,27%	30,94%	6,73%	0,71%	86,82%
Memcpy C	60,25%	63,73%	59,83%	59,34%	73,52%
Kernel	-1,11%	14,11%	2,09%	0,28%	82,39%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	71,64%	-32,31%	0,00%	40,32%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

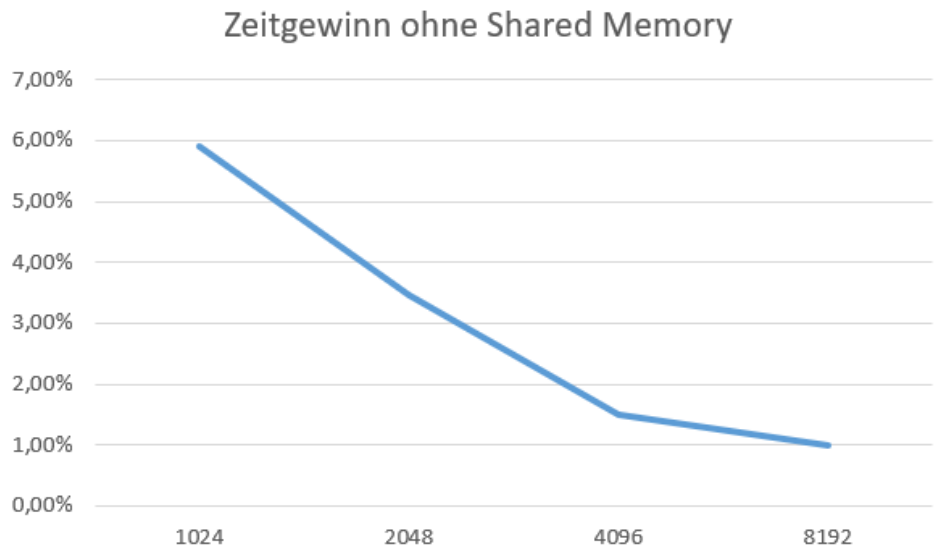
4096	Min	Max	Avg	Median	StdDV
time*	4,67%	8,32%	6,27%	6,15%	46,72%
GFLOPS	-9,08%	-4,90%	-6,64%	-6,55%	39,15%
Memcpy AB	-1,70%	9,69%	1,06%	-0,57%	30,96%
Memcpy C	58,51%	60,25%	60,48%	60,89%	83,66%
Kernel	-0,29%	1,15%	0,47%	-0,66%	49,56%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,34%	67,01%	31,13%	4,17%	63,59%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

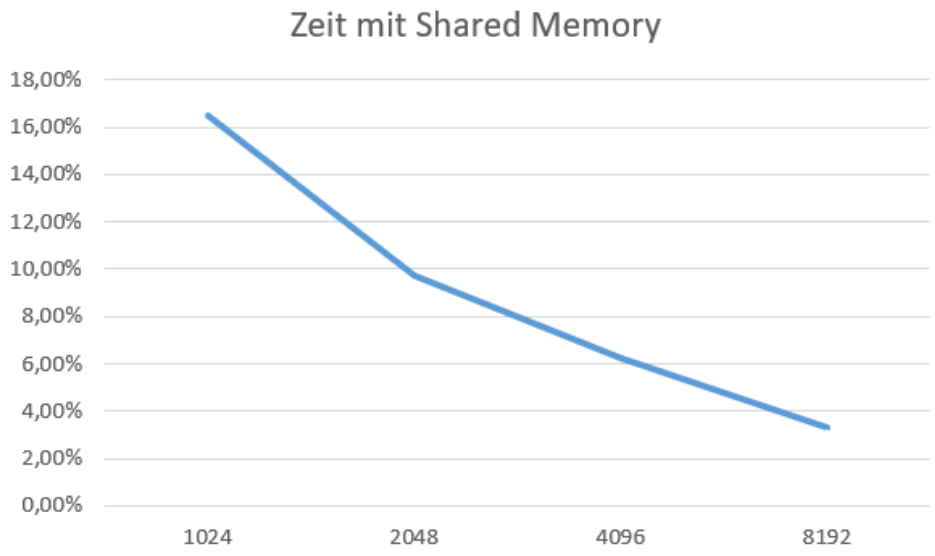
2048	Min	Max	Avg	Median	StdDV
time*	8,41%	13,32%	9,71%	8,80%	43,60%
GFLOPS	-15,36%	-9,19%	-10,64%	-9,65%	29,38%
Memcpy AB	0,24%	12,77%	2,52%	2,16%	33,03%
Memcpy C	58,52%	73,96%	61,69%	59,30%	95,79%
Kernel	-0,04%	-3,46%	-0,39%	-0,12%	-87,98%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	13,93%	-41,48%	0,00%	21,97%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

8192	Min	Max	Avg	Median	StdDV
time*	3,09%	4,71%	3,31%	3,21%	61,34%
GFLOPS	-4,95%	-3,18%	-3,42%	-3,31%	58,25%
Memcpy AB	0,39%	17,18%	3,53%	2,45%	66,98%
Memcpy C	57,99%	62,95%	60,12%	60,12%	96,54%
Kernel	-0,10%	-0,32%	-0,11%	-0,09%	-27,21%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	5,07%	20,00%	26,25%	10,43%	38,05%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

Auch hier trägt die Reduktion in Memcpy für C im Wesentlichen zum Leistungsgewinn von jeweils ca. 60 % bei. Mit zunehmender Größe fällt der Leistungsgewinn insgesamt ab, beginnend von 16,5 % Zeit (GFLOPS 20 %) bis auf ca. 3 % (Zeit und GFLOPS). Die Auswirkungen auf die Kernelzeit sind unwesentlich, ebenso die Veränderungen bei den übrigen Stationen.

Im Ergebnis zeigt sich die optimierende Wirkung von CudaMallocHost hinreichend deutlich.





Verwendet wurden die Average-Werte.

Die Abnahme im Zeitgewinn erklärt sich daraus, dass mit zunehmender Matrixgröße der lineare Performance-Gewinn in der Übertragung hinter der höherkomplexen Berechnung (Matrixmultiplikation: grundsätzlich $O(n^3)$, mit Parallelität in den hier verwendeten Modellen auf bis zu $O(n^2)$ reduzierbar) zurückfällt. Bei Shared Memory wirkt dieser noch zeitersparend mit einem Faktor von ca. 3. Der Verlauf ist jedoch hinreichend ähnlich, der Betrag der negativen Steigung nimmt jedoch etwas früher ab.

Performance-Vergleich Shared-Memory mit Malloc

1024	Min	Max	Avg	Median	StdDV
time*	51,35%	44,90%	50,21%	50,44%	24,20%
GFLOPS	-81,49%	-105,56%	-101,23%	-101,77%	-191,99%
Memcpy AB	1,32%	15,03%	-1,60%	-0,78%	27,44%
Memcpy C	-1,21%	3,29%	0,11%	0,39%	8,50%
Kernel	69,31%	65,37%	68,53%	68,81%	33,40%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	-215,90%	19,38%	100,00%	-51,76%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

4096	Min	Max	Avg	Median	StdDV
time*	73,64%	71,59%	72,96%	73,13%	-37,64%
GFLOPS	-252,02%	-279,30%	-270,05%	-272,22%	-1742,32%
Memcpy AB	-0,87%	-14,49%	-0,75%	0,56%	-36,24%
Memcpy C	0,70%	4,01%	-0,16%	-2,64%	7,89%
Kernel	78,58%	77,05%	78,24%	78,52%	-27,86%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	23,00%	-170,69%	-36,06%	-2,13%	-142,31%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

2048	Min	Max	Avg	Median	StdDV
time*	62,49%	57,83%	61,53%	62,19%	-94,10%
GFLOPS	-137,11%	-166,62%	-160,40%	-164,48%	-1145,51%
Memcpy AB	0,15%	-5,02%	-1,06%	0,31%	-35,79%
Memcpy C	2,10%	0,43%	0,46%	3,77%	-10,96%
Kernel	72,20%	72,14%	72,27%	72,44%	57,88%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	9,06%	59,40%	100,00%	13,96%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

8192	Min	Max	Avg	Median	StdDV
time*	75,22%	74,74%	75,15%	75,18%	22,29%
GFLOPS	-295,84%	-303,57%	-302,48%	-302,91%	-1144,07%
Memcpy AB	0,05%	-16,71%	-2,50%	-0,86%	-160,65%
Memcpy C	0,41%	-3,92%	-2,41%	-3,26%	-65,17%
Kernel	77,73%	77,71%	77,76%	77,75%	80,54%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	-28,42%	-15,31%	-42,68%	-10,85%	-80,73%
Free	0,00%	100,00%	100,00%	0,00%	100,00%

Durch shared konnten deutlich Gewinne erzielt werden, welche mit zunehmender Größe zunahmen, jedoch in den Matrixgrößen von 4096 zu 8192 keine bedeutende Steigerung mehr aufwiesen bzw. beim Kernel sogar ein leichter Rückgang. Die Werte im Speicherübertragungen sind unwesentlich, die übrigen Werte sind hier ebenfalls aussagelos.

Performance-Vergleich Shared-Memory mit CudaMallocHost

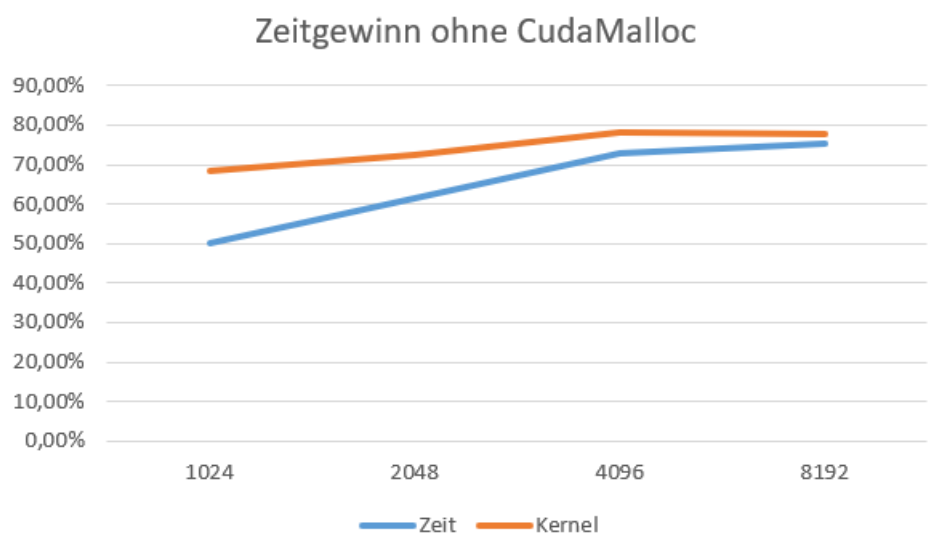
1024	Min	Max	Avg	Median	StdDV
time*	55,12%	57,44%	55,82%	55,67%	83,84%
GFLOPS	-134,97%	-122,79%	-126,11%	-125,60%	15,82%
Memcpy AB	0,42%	32,93%	6,87%	1,00%	88,56%
Memcpy C	-1,47%	-12,96%	0,05%	1,36%	-39,72%
Kernel	69,00%	70,91%	69,38%	69,14%	90,04%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	80,39%	12,63%	0,00%	62,23%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

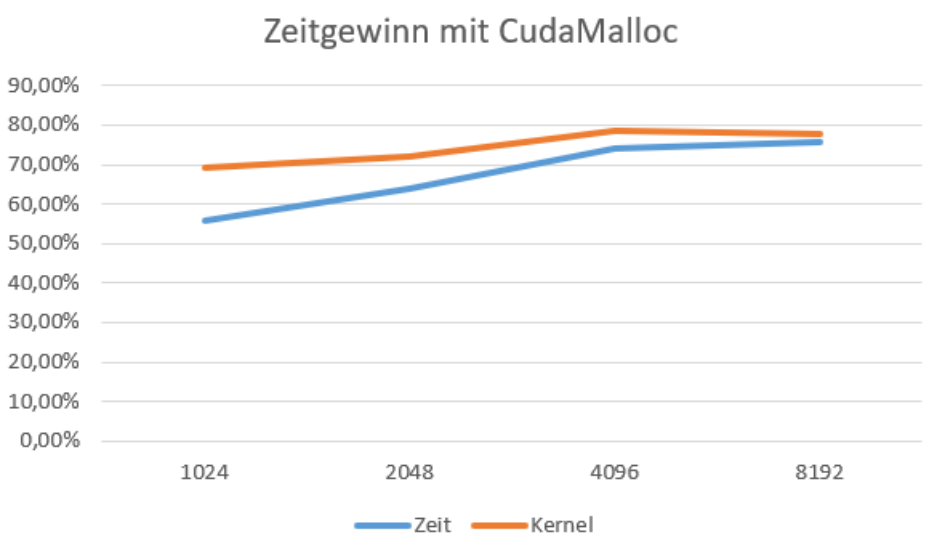
4096	Min	Max	Avg	Median	StdDV
time*	74,46%	73,60%	74,27%	74,40%	33,57%
GFLOPS	-278,74%	-291,47%	-288,68%	-290,62%	-887,33%
Memcpy AB	0,28%	-6,11%	0,78%	1,86%	-29,70%
Memcpy C	0,17%	-4,37%	-0,08%	0,21%	-97,52%
Kernel	78,51%	77,33%	78,35%	78,37%	48,88%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	14,16%	14,80%	9,48%	5,69%	15,76%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

2048	Min	Max	Avg	Median	StdDV
time*	64,40%	62,75%	64,02%	64,21%	20,77%
GFLOPS	-168,49%	-180,89%	-178,08%	-179,44%	-500,80%
Memcpy AB	0,04%	15,19%	0,89%	-0,96%	34,90%
Memcpy C	-0,53%	15,14%	5,75%	5,32%	78,09%
Kernel	72,21%	71,08%	72,18%	72,41%	-1,44%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	0,00%	43,64%	51,96%	31,78%	37,25%
Free	0,00%	0,00%	0,00%	0,00%	0,00%

8192	Min	Max	Avg	Median	StdDV
time*	75,76%	75,66%	75,73%	75,75%	59,58%
GFLOPS	-310,77%	-312,57%	-312,10%	-312,44%	-585,28%
Memcpy AB	-0,03%	0,44%	0,11%	-0,61%	14,00%
Memcpy C	0,12%	-0,44%	0,02%	0,04%	-29,09%
Kernel	77,68%	77,55%	77,68%	77,70%	67,24%
Malloc	0,00%	0,00%	0,00%	0,00%	0,00%
Sync	-2,30%	-0,40%	-2,38%	-1,20%	-11,99%
Free	0,00%	-9,17%	-219,91%	0,00%	-76,86%

Auch hier konnte durch shared eine deutliche Leistungssteigerung mit ähnlichen Ausprägungen erreicht werden. Eine Tendenz einer Gewinnzunahme zeigte sich hier wie oben mit einem Knick von 4096 auf 8192. Der wesentliche Anteil macht der Zugewinn direkt im Kernel aus. Die Werte im Speicherübertragungen sind auch hier unwesentlich, die übrigen Werte sind ebenfalls aussagelos.





Verwendet wurden die Average-Werte. Die Zeitvorteile nehmen zu. Dies dürfte damit zusammenhängen, dass die eigentliche Arbeit in der höherkomplexen Matrixmultiplikation stattfindet. Das leichte Abfallen im Kernel von 4096 zu 8192 ist insoweit gegenläufig.

Anlage 1 Zusammengefasste Werte

Zusammenfassungen aus Anlage 4

<u>Performance-Werte</u> (time* bezeichnet die Zeitmessung aus der Vorlage) **ohne** shared-Memory

Simple **Host**

1024	Min	Max	Avg	Median	StdDV
time*	7,9695	8,84365	8,2775116	8,23094	0,275462688
GFLOPS	121,414	134,731	129,85796	130,452	4,210520808
Memcpy AB	1,2057	1,92874	1,3412284	1,26586	0,170103734
Memcpy C	0,35936	0,425056	0,39341952	0,39232	0,017256503
Kernel	6,35267	6,99056	6,5184732	6,45763	0,179722812
Malloc	0	0	0	0	0
Sync	0	0,050592	0,00649728	0	0,012280469
Free	0	0	0	0	0

2048	Min	Max	Avg	Median	StdDV
time*	57,5964	60,624	58,370568	58,1321	0,705052269
GFLOPS	141,692	149,14	147,18316	147,766	1,744907601
Memcpy AB	4,96218	7,4663	5,3230076	5,01981	0,609615009
Memcpy C	1,32128	1,67082	1,4354972	1,42275	0,095057438
Kernel	51,0103	52,1054	51,578532	51,5711	0,254581774
Malloc	0	0	0	0	0
Sync	0	0,050528	0,01966592	0,013696	0,011552711
Free	0	0	0	0	0

4096	Min	Max	Avg	Median	StdDV
time*	506,859	521,217	509,90148	508,921	3,019611049
GFLOPS	131,844	135,579	134,77476	135,03	0,786349301
Memcpy AB	20,0248	25,0946	21,455296	21,2568	1,107443334
Memcpy C	5,16403	5,4111	5,2615456	5,256	0,049764989

Simple Malloc

1024	Min	Max	Avg	Median	StdDV
time*	8,54682	10,0484	8,7973168	8,66202	0,369273484
GFLOPS	106,857	125,631	122,24848	123,96	4,653007072
Memcpy AB	1,21338	2,20432	1,318218	1,25242	0,203478401
Memcpy C	0,906272	1,36867	0,97998696	0,955488	0,099489998
Kernel	6,34646	6,83626	6,478102	6,4064	0,15266868
Malloc	0	0	0	0	0
Sync	0	0,011072	0,00532224	0,009856	0,005122061
Free	0	0	0	0	0

2048	Min	Max	Avg	Median	StdDV
time*	59,6924	61,7639	60,470584	60,3257	0,510244536
GFLOPS	139,077	143,903	142,06156	142,393	1,191885568
Memcpy AB	4,97971	6,91194	5,3551144	5,19539	0,436427909
Memcpy C	3,2705	5,4681	3,5484208	3,43958	0,446091195
Kernel	50,9673	52,2811	51,536856	51,5544	0,326201475
Malloc	0	0	0	0	0
Sync	0	0,036384	0,01644928	0,011872	0,010798253
Free	0	0	0	0	0

4096	Min	Max	Avg	Median	StdDV
time*	515,145	528,439	517,69092	516,736	2,735290082
GFLOPS	130,042	133,398	132,74596	132,988	0,692603291
Memcpy AB	19,4637	25,7523	21,356536	20,8605	1,52698667
Memcpy C	12,5129	14,8018	13,303728	13,0653	0,652983144

Kernel	481,335	495,211	483,14136	482,352	3,113073502
Malloc	0	0	0	0	0
Sync	0,011072	0,039136	0,01768704	0,014048	0,00829655
Free	0	0	0	0	0

Kernel	481,572	494,838	483,0028	482,581	2,467475374
Malloc	0	0	0	0	0
Sync	0,012384	0,037344	0,01708672	0,013536	0,007922411
Free	0	0	0	0	0

8192	Min	Max	Avg	Median	StdDV
time*	3945,25	3964,86	3955,2328	3956,14	5,255462316
GFLOPS	138,657	139,346	138,99476	138,963	0,184692237
Memcpy AB	76,8313	82,7437	78,670272	77,8856	1,850850315
Memcpy C	20,4359	20,6283	20,54494	20,5356	0,04929447
Kernel	3846,85	3866,24	3855,9776	3857,64	5,223605866
Malloc	0	0	0	0	0
Sync	0,011136	0,032256	0,01561856	0,013312	0,006025232
Free	0	36,7902	1,471608	0	7,209377403

8192	Min	Max	Avg	Median	StdDV
time*	3982,05	4009,7	3995,1108	3992,79	7,072001652
GFLOPS	137,106	138,059	137,60756	137,687	0,24369474
Memcpy AB	77,1926	85,2304	79,467156	79,6447	1,849406781
Memcpy C	48,7881	53,815	50,292948	49,8421	1,11202297
Kernel	3852,44	3881,27	3865,3268	3863,26	6,911455835
Malloc	0	0	0	0	0
Sync	0,009344	0,035104	0,01519488	0,013568	0,006027398
Free	0	26,9029	1,076116	0	5,271870208

<u>Performance-Werte</u> (time* bezeichnet die Zeitmessung aus der Vorlage) **mit** shared-Memory

Shared Host

1024	Min	Max	Avg	Median	StdDV
time*	3,57709	3,76378	3,6573508	3,64845	0,044514874
GFLOPS	285,283	300,172	293,62784	294,301	3,54447273
Memcpy AB	1,20064	1,29354	1,2491036	1,25325	0,019464737
Memcpy C	0,36464	0,480128	0,39323904	0,386976	0,02411022
Kernel	1,96963	2,03325	1,995918	1,99277	0,0179026
Malloc	0	0	0	0	0
Sync	0	0,00992	0,0056768	0,009248	0,004638636
Free	0	0	0	0	0

Shared Malloc

1024	Min	Max	Avg	Median	StdDV
time*	4,15789	5,53658	4,3801448	4,29302	0,279895875
GFLOPS	193,936	258,242	246,00172	250,113	13,58648163
Memcpy AB	1,19738	1,87312	1,3392492	1,26221	0,147634621
Memcpy C	0,91728	1,32362	0,97891384	0,951744	0,091034358
Kernel	1,948	2,3672	2,0385172	1,99834	0,101684362
Malloc	0	0	0	0	0
Sync	0	0,034976	0,00429056	0	0,007773073
Free	0	0	0	0	0

2048	Min	Max	Avg	Median	StdDV
time*	20,5047	22,5798	21,00188	20,8028	0,558626127
GFLOPS	380,426	418,925	409,28668	412,922	10,48335964

2048	Min	Max	Avg	Median	StdDV
time*	22,3883	26,0487	23,260036	22,8093	0,990395958
GFLOPS	329,765	383,679	369,93188	376,598	14,84501271

Memcpy AB	4,96038	6,33187	5,2754828	5,06781	0,396866796
Memcpy C	1,32826	1,41782	1,3529952	1,34707	0,020827221
Kernel	14,174	15,0701	14,3479	14,2276	0,258252224
Malloc	0	0	0	0	0
Sync	0	0,02848	0,00944768	0,009344	0,007249792
Free	0	0	0	0	0

Memcpy AB	4,97226	7,25891	5,4119724	5,17952	0,592618165
Memcpy C	3,20195	5,44474	3,5320632	3,30979	0,49500492
Kernel	14,169	14,5666	14,292832	14,2099	0,137384703
Malloc	0	0	0	0	0
Sync	0	0,033088	0,00667776	0	0,009291062
Free	0	0	0	0	0

4096	Min	Max	Avg	Median	StdDV
time*	129,475	137,619	131,21436	130,286	2,00591174
GFLOPS	499,346	530,757	523,8378	527,451	7,763880293
Memcpy AB	19,9682	26,6268	21,288444	20,8619	1,43640647
Memcpy C	5,15546	5,64749	5,2655956	5,24522	0,098294077
Kernel	103,449	112,279	104,62364	104,329	1,591258468
Malloc	0	0	0	0	0
Sync	0,009504	0,033344	0,01601024	0,013248	0,00698908
Free	0	0	0	0	0

4096	Min	Max	Avg	Median	StdDV
time*	135,816	150,116	139,99216	138,823	3,764821342
GFLOPS	457,775	505,975	491,22384	495,014	12,75996324
Memcpy AB	19,6337	29,4847	21,517256	20,7439	2,080440651
Memcpy C	12,4258	14,2086	13,325216	13,4102	0,601460087
Kernel	103,152	113,584	105,11664	103,644	3,154904599
Malloc	0	0	0	0	0
Sync	0,009536	0,101088	0,02324864	0,013824	0,019196687
Free	0	0	0	0	0

8192	Min	Max	Avg	Median	StdDV
time*	956,253	965,238	959,76692	959,208	2,124512385
GFLOPS	569,555	574,906	572,80412	573,135	1,265667992
Memcpy AB	76,8539	82,3763	78,580764	78,3627	1,591657074
Memcpy C	20,4108	20,7192	20,540012	20,5273	0,063635854
Kernel	858,634	867,865	860,60612	860,342	1,711226305
Malloc	0	0	0	0	0
Sync	0,011392	0,032384	0,01598976	0,013472	0,006747911
Free	0	40,1623	4,70778	0	12,75079258

8192	Min	Max	Avg	Median	StdDV
time*	986,696	1012,97	992,65364	990,993	5,495888622
GFLOPS	542,714	557,168	553,84104	554,752	3,031736512
Memcpy AB	77,1562	99,4691	81,455704	80,3295	4,820455076
Memcpy C	48,5895	55,9249	51,50348	51,4688	1,83675841
Kernel	857,776	865,089	859,66544	859,578	1,345219687
Malloc	0	0	0	0	0
Sync	0,012	0,04048	0,02168064	0,01504	0,010893353
Free	0	0	0	0	0

Anlage 2 Zeitmessung

Programmergänzungen zur Zeitmessung (hierfür wurde zur Vereinfachung C++ statt C verwendet) am Beispiel für shared CudaMallocHost, für die Zeitmessung relevante Ergänzungen blau gefärbt.

```
#include <stdio.h>

#include <stdlib.h>

#include <assert.h>

#include <cuda.h>

#include <fstream>

// Thread block size: BLOCK_SIZE * BLOCK_SIZE

#define BLOCK_SIZE  16

#define __N__ 8192

// In Anlehnung an http://www.techdarting.com/2014/03/matrix-multiplication-in-cuda-using.html

__global__ void dgemm_gpu_shared(const float* a, const float* b, float* c, const int n) {

        //Todo: implement Kernel Here

        __shared__ float sA[BLOCK_SIZE][BLOCK_SIZE];   // Tile size of 32x32

        __shared__ float sB[BLOCK_SIZE][BLOCK_SIZE];

        int Row = blockDim.y * blockIdx.y + threadIdx.y;

        int Col = blockDim.x * blockIdx.x + threadIdx.x;

        float Cvalue = 0.0;

        sA[threadIdx.y][threadIdx.x] = 0.0;
```

```
        sB[threadIdx.y][threadIdx.x] = 0.0;
        int Row_n = Row * n;
        int anz_BL = (n + BLOCK_SIZE - 1) / BLOCK_SIZE; // Number of Blocks in Matrix
        for (int ph = 0; ph < anz_BL; ph++) {
                // Load
                int ph_BLOCK_SIZE = ph * BLOCK_SIZE;
                sA[threadIdx.y][threadIdx.x] = a[Row_n + threadIdx.x + ph_BLOCK_SIZE];
                sB[threadIdx.y][threadIdx.x] = b[(threadIdx.y + ph_BLOCK_SIZE) * n + Col];
                __syncthreads();
                // Calc
                for (int j = 0; j < BLOCK_SIZE; ++j) {
                        Cvalue += sA[threadIdx.y][j] * sB[j][threadIdx.x];
                }
        }
        //Store
        c[Row_n + Col] = Cvalue;
}
// get compute performance
float getGflops(int width, float time) {
        float gf = (1.0e-6 * width * width * width / time);
```

```
        return gf;
}
int main(int argc, const char** argv) {
        int n = __N__; // dimension of square matrices
        float* h_a = 0, * h_b = 0, * h_c = 0;
        float* d_a, * d_b, * d_c;
        int row, col;
        float absError, maxAbsError = 0.0, sumAbsError = 0.0;
        size_t size;
        float time, timeFree, timeMemcpy, timeMemcpyC, timeMalloc, timeKernel, timeSync;
        cudaEvent_t start, stop;
        clock_t tstart, tfinish;
        double intervall;
        cudaEvent_t startFree, stopFree;
        cudaEvent_t startMemcpy, stopMemcpy, startMemcpyC, stopMemcpyC;
        cudaEvent_t startMalloc, stopMalloc;
        cudaEvent_t startKernel, stopKernel;
        cudaEvent_t startSync, stopSync;
        //Möglichkeit mit GPUs Zeit zu messen
        cudaEventCreate(&start);
```

```
cudaEventCreate(&stop);
cudaEventCreate(&startFree);
cudaEventCreate(&stopFree);
cudaEventCreate(&startMemcpy);
cudaEventCreate(&stopMemcpy);
cudaEventCreate(&startMemcpyC);
cudaEventCreate(&stopMemcpyC);
cudaEventCreate(&startMalloc);
cudaEventCreate(&stopMalloc);
cudaEventCreate(&startKernel);
cudaEventCreate(&stopKernel);
cudaEventCreate(&startSync);
cudaEventCreate(&stopSync);
// Größe der Matrix kann als Argument  übergeben werden
if (argc > 1) {
        n = atoi(argv[1]);
}
size = n * n * sizeof(float);
//TODO  Allokiere Speicher auf dem host und GPU
cudaEventRecord(startMalloc, 0);
```

```
h_a = (float*)malloc(size);
h_b = (float*)malloc(size);
//h_c = (float*)malloc(size);
//cudaMallocHost((void**)&h_a, size);
//cudaMallocHost((void**)&h_b, size);
cudaMallocHost((void**)&h_c, size);
//tstart = clock();
cudaMalloc(&d_a, size);
cudaMalloc(&d_b, size);
cudaMalloc(&d_c, size);
//tfinish = tstart - clock();
//intervall = tfinish / (double)CLOCKS_PER_SEC;
cudaEventRecord(stopMalloc, 0);
cudaEventElapsedTime(&timeMalloc, startMalloc, stopMalloc);
cudaEventDestroy(startMalloc);
cudaEventDestroy(stopMalloc);
//Initialisierung, parallelisiert mit OpenMP
if (h_a && h_b) {
        //#pragma omp parallel for
        for (row = 0; row < n; row++) {
```

```
                for (col = 0; col < n; col++) {
                        h_a[row * n + col] = (row == col) ? 1.0 : 0.0;
                        h_b[row * n + col] = row * n + col;
                }
        }
}
else return;
// Führe die Matrix-Matrix Multiplikation aus
//16x16 ist vorgegeben
dim3 blockDim(BLOCK_SIZE, BLOCK_SIZE);
dim3 gridDim((n + BLOCK_SIZE - 1) / blockDim.x, (n + BLOCK_SIZE - 1) / blockDim.y);
cudaEventRecord(start, 0);
//TODO kopiere daten auf die GPU und Rufe den Kernel auf  und Kopiere das Ergebniss zurrück
cudaEventRecord(startMemcpy, 0);
cudaMemcpy(d_a, h_a, size, cudaMemcpyHostToDevice);
cudaMemcpy(d_b, h_b, size, cudaMemcpyHostToDevice);
cudaEventRecord(stopMemcpy, 0);
cudaEventRecord(startKernel, 0);
dgemm_gpu_shared << < gridDim, blockDim >> > (d_a, d_b, d_c, n);
cudaEventRecord(stopKernel, 0);
```

```
cudaEventRecord(startMemcpyC, 0);
cudaMemcpy(h_c, d_c, size, cudaMemcpyDeviceToHost);
cudaEventRecord(stopMemcpyC, 0);
cudaEventRecord(startSync, 0);
cudaDeviceSynchronize();
cudaEventRecord(stopSync, 0);
cudaEventRecord(stop, 0);
cudaEventElapsedTime(&timeMemcpy, startMemcpy, stopMemcpy);
cudaEventDestroy(startMemcpy);
cudaEventDestroy(stopMemcpy);
cudaEventElapsedTime(&timeMemcpyC, startMemcpyC, stopMemcpyC);
cudaEventDestroy(startMemcpyC);
cudaEventDestroy(stopMemcpyC);
cudaEventElapsedTime(&timeKernel, startKernel, stopKernel);
cudaEventDestroy(startKernel);
cudaEventDestroy(stopKernel);
cudaEventElapsedTime(&timeSync, startSync, stopSync);
cudaEventDestroy(startSync);
cudaEventDestroy(stopSync);
// Teste das Ergebnis
```

```
for (row = 0; row < n; ++row) {
        for (col = 0; col < n; ++col) {
                absError = fabs(h_c[row * n + col] - h_b[row * n + col]);
                sumAbsError += absError;
                if (absError > maxAbsError)
                        maxAbsError = absError;
        }
}
cudaEventElapsedTime(&time, start, stop);
cudaEventDestroy(start);
cudaEventDestroy(stop);
//TODO Gebe den Speicher auf GPU und Host frei
cudaEventRecord(startFree, 0);
cudaFree(d_c);
cudaFree(d_b);
cudaFree(d_a);
//free(h_c);
free(h_b);
free(h_a);
cudaFreeHost(h_c);
```

```
//cudaFreeHost(h_b);
//cudaFreeHost(h_a);
cudaEventRecord(stopFree, 0);
cudaEventElapsedTime(&timeFree, startFree, stopFree);
cudaEventDestroy(startFree);
cudaEventDestroy(stopFree);
printf("\nmaxAbsError: %4.4f, sumAbsError: %4.4f\n", maxAbsError, sumAbsError);
if (maxAbsError < 2.0e-5) {
	printf("\nProgram terminated SUCCESSFULLY.\n");
	printf("\nKernel Execution Time: %f ms (dim C: %d * %d)", time, n, n);
	printf("\nThis corresponds to: %4.4f GFLOPS\n\n", getGflops(n, time));
}
else {
	printf("\n--> Result not correct:  check your code\n\n");
}
std::ofstream file("C:/Users/nbaum/Documents/fu_hagen/parallel/Aufgabe 2/Zeitmessung/simple1024.dat", std::ios::out | std::ios::app);
if (file.fail()) {
	printf("\Schreibfehler Datei\n"); return 0;
}
file << time << std::endl;
```

```
        file << getGflops(n, time) << std::endl;
        file << timeMemcpy << std::endl;
        file << timeMemcpyC << std::endl;
        file << timeKernel << std::endl;
        file << timeMalloc << std::endl;
        file << timeSync << std::endl;
        file << timeFree << std::endl;
        file << "******\n";
        file.close();
        return 0;
}
```

Anlage 3 Kernelvarianten

3.1 Im Kernel wird nicht nur eine C-Zelle, sondern ein Block von C-Zellen befüllt.

```
// Thread block size: BLOCK_SIZE * BLOCK_SIZE
#define BLOCK_SIZE 16
// number of partial blocks
#define PART_BLOCKS
__global__ void dgemm_gpu_shared(const float* a, const float* b, float* c, const int n, const int by, const int bx, const int anzahl_bl, const int m_block_x, const int m_block_y, const int m_block_z) {
```

```
//Todo: implement Kernel Here
int x = threadIdx.x + blockIdx.x * blockDim.x;
int y = threadIdx.y + blockIdx.y * blockDim.y;
int id = x + y * blockDim.x * gridDim.x;
__shared__ float shared_a[BLOCK_SIZE * BLOCK_SIZE];
__shared__ float shared_b[BLOCK_SIZE * BLOCK_SIZE];
__shared__ float shared_c[BLOCK_SIZE * BLOCK_SIZE];
if (y >= m_block_y)return;
if (x >= m_block_x)return;
int cid = bx + by * n;
int aid, bid;
int ix = x;
int iy = y;
int t_id = threadIdx.x + threadIdx.y * blockDim.x;
for (int imblock = 0; imblock < anzahl_bl; ++imblock)
        //for (int iy = 0; iy < M_BLOCK_Y; ++iy)
                //for (int ix = 0; ix < M_BLOCK_X; ++ix)
                        for (int i = 0; i < m_block_z; ++i) {
                                aid = n * by + m_block_x * imblock; // starting point
                                aid += iy * n + i; // extend in partial matrix
```

```
                    bid = bx + m_block_y * imblock * n; // starting point
                    bid += ix + i * n; // extend in partial matrix
                    //c[cid + ix + n * iy] += a[aid] * b[bid];
                    shared_a[t_id] = a[aid];
                    shared_b[t_id] = b[bid];
                    shared_c[t_id] = shared_a[t_id] * shared_b[t_id];
                    c[cid + ix + n * iy] += shared_c[t_id];
                }
}
```

Aufruf des Kernels in main:

```
int m_block_x, m_block_y, m_block_z; // dimension of partial matrix
m_block_x = m_block_y = m_block_z = n / PART_BLOCKS;
    // dimension smaller thanbBlock_size is out of scope
    if (n < BLOCK_SIZE||n<m_block_x||n<m_block_y||n<m_block_z) {
        printf("invalid parameters\n");
        return -1;
    }
    // number of partial martices
    int anz_m_bl_x = n / m_block_x;
    int anz_m_bl_y = n / m_block_y;
```

```
int anz_m_bl_z = n / m_block_z;
#pragma omp parallel for
for (int y = 0; y < anz_m_bl_y; ++y) {
    for (int x = 0; x < anz_m_bl_x; ++x) {
        dgemm_gpu_shared << < gridDim, blockDim >> > (d_a, d_b, d_c, n, y*m_block_y, x*m_block_x,
            anz_m_bl_y, m_block_x, m_block_y, m_block_z);
    }
}
```

3.2 Ähnlich wie die Ausgangsvariante, jedoch Verwendung von Shared-Memory auch für C

```
_global__ void dgemm_gpu_shared_(const float* a, const float* b, float* c, const int n) {
    //Todo: implement Kernel Here
    int x = threadIdx.x + blockIdx.x * blockDim.x;
    int y = threadIdx.y + blockIdx.y * blockDim.y;
    int id = x + y * blockDim.x * gridDim.x;
    int numb_bl = n / BLOCK_SIZE; // increment
    int tx = threadIdx.x;
    int ty = threadIdx.y;
    __shared__ float shared_A[BLOCK_SIZE][BLOCK_SIZE];
    __shared__ float shared_B[BLOCK_SIZE][BLOCK_SIZE];
    __shared__ float shared_C[BLOCK_SIZE][BLOCK_SIZE];
```

```
        int vx = x;
        int vy = id - vx; // Zeile
        shared_C[tx][ty] = 0;
        for (int j = 0; j < numb_bl; ++j) {
                //Load
                shared_A[tx][ty] = a[vy + tx + j * BLOCK_SIZE];
                shared_B[tx][ty] = b[vx + ty * n  + j * BLOCK_SIZE * n];
                __syncthreads();

                // Calc
                for (int i = 0; i < BLOCK_SIZE; ++i)
                        shared_C[tx][ty] += shared_A[i][ty] * shared_B[tx][i];
        }
        //Store
        c[id] += shared_C[tx][ty];
}
```

Anlage 4 Messeinzeldaten

Es wurden pro Matrixgröße, pro mit/ohne shared, pro CudaMallocHost/Malloc jeweils 25 Versuche durchgeführt. Vor Durchführung wurden jeweils drei Durchläufe durchgeführt, um Initialisierungsverzögerungen zu vermeiden.

1024,00000	simple	Host										
time*	8,257	8,456	8,588	8,414	8,273	7,984	8,178	8,088	8,834	8,018	8,049	8,008
GFLOPS	130,038	126,977	125,026	127,616	129,793	134,494	131,289	132,754	121,548	133,919	133,401	134,090
Memcpy AB	1,312	1,266	1,396	1,228	1,363	1,222	1,315	1,263	1,779	1,227	1,255	1,207
Memcpy C	0,404	0,407	0,425	0,398	0,411	0,388	0,387	0,386	0,408	0,388	0,420	0,418
Kernel	6,482	6,760	6,748	6,763	6,480	6,355	6,458	6,419	6,627	6,384	6,355	6,361
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,051	0,012	0,010	-	-	-	-	-	-	0,010	-	-
Free	-	-	-	-	-	-	-	-	-	-	-	-
8,232	8,844	8,150	8,407	8,756	8,231	7,993	8,280	8,075	7,970	8,739	8,018	8,097
130,438	121,414	131,740	127,724	122,631	130,452	134,328	129,676	132,975	134,731	122,874	133,909	132,612
1,406	1,929	1,260	1,238	1,335	1,431	1,223	1,377	1,254	1,206	1,488	1,232	1,321
0,374	0,361	0,369	0,399	0,409	0,359	0,385	0,376	0,390	0,392	0,379	0,403	0,398
6,413	6,532	6,500	6,750	6,991	6,418	6,364	6,490	6,411	6,353	6,833	6,358	6,358
-	-	-	-	-	-	-	-	-	-	-	-	-
-	0,010	-	-	0,012	-	-	0,029	-	-	0,029	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
******	******	******	******	******	******	******	******	******	******	******	******	******
2048,00000	simple	Host										
time*	58,247	58,238	57,939	57,596	58,494	58,444	57,632	59,149	57,687	59,637	57,922	58,132
GFLOPS	147,475	147,497	148,259	149,140	146,851	146,977	149,048	145,226	148,905	144,037	148,303	147,767
Memcpy AB	4,962	5,226	4,987	5,020	4,998	5,466	5,078	5,650	4,992	6,396	5,006	4,969
Memcpy C	1,671	1,423	1,394	1,351	1,359	1,650	1,523	1,491	1,335	1,340	1,372	1,554

Kernel	51,586	51,566	51,494	51,184	52,105	51,277	51,010	51,959	51,320	51,860	51,519	51,571
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,010	0,014	0,051	0,031	0,013	0,038	0,011	0,011	0,028	0,030	0,010	0,013
Free	-	-	-	-	-	-	-	-	-	-	-	-
57,822	58,132	58,087	57,812	58,161	58,514	59,599	58,030	58,051	57,880	60,624	58,772	58,665
148,558	147,766	147,881	148,584	147,692	146,802	144,129	148,026	147,973	148,409	141,692	146,158	146,424
5,243	4,982	5,014	4,977	4,988	5,290	6,668	5,052	4,986	4,970	7,466	5,343	5,347
1,376	1,514	1,355	1,321	1,504	1,468	1,347	1,453	1,347	1,427	1,363	1,511	1,438
51,158	51,612	51,673	51,489	51,647	51,732	51,539	51,500	51,677	51,458	51,776	51,895	51,855
-	-	-	-	-	-	-	-	-	-	-	-	-
0,030	0,015	0,034	0,014	-	0,010	0,030	0,016	0,031	0,011	0,011	0,013	0,017
-	-	-	-	-	-	-	-	-	-	-	-	-
******	******	******	******	******	******	******	******	******	******	******	******	******
4096,00000	simple	Host										
time*	517,133	521,217	510,000	508,626	508,147	508,364	509,843	508,494	507,522	508,639	509,765	506,859
GFLOPS	132,885	131,844	134,744	135,108	135,235	135,178	134,786	135,143	135,402	135,105	134,806	135,579
Memcpy AB	20,229	20,740	22,686	20,882	21,551	21,519	22,150	21,656	20,377	21,568	22,675	20,025
Memcpy C	5,258	5,243	5,261	5,263	5,229	5,248	5,282	5,256	5,276	5,164	5,261	5,253
Kernel	491,622	495,211	482,022	482,408	481,335	481,574	482,355	481,552	481,723	481,857	481,797	481,553
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,013	0,012	0,012	0,031	0,015	0,011	0,013	0,011	0,014	0,032	0,017	0,015
Free	-	-	-	-	-	-	-	-	-	-	-	-
512,589	509,881	510,474	508,797	510,657	508,921	509,100	509,894	507,922	508,277	508,775	509,496	508,145
134,063	134,776	134,619	135,063	134,571	135,030	134,982	134,772	135,295	135,201	135,069	134,877	135,236
25,095	21,735	22,417	20,718	22,729	20,247	21,252	22,280	20,667	20,639	21,257	21,048	20,245
5,213	5,248	5,166	5,269	5,243	5,314	5,306	5,217	5,256	5,323	5,411	5,246	5,331
482,203	482,871	482,796	482,775	482,650	483,285	482,519	482,352	481,977	482,289	482,084	483,179	482,545

-	-	-	-	-	-	-	-	-	-	-	-	-
0,039	0,012	0,033	0,015	0,017	0,032	0,012	0,021	0,013	0,012	0,012	0,012	0,016
-	-	-	-	-	-	-	-	-	-	-	-	-
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8192,00000	simple	Host										
time*	3.956,140	3.945,250	3.945,680	3.948,010	3.950,950	3.951,090	3.962,300	3.962,650	3.956,200	3.959,810	3.951,700	3.947,640
GFLOPS	138,963	139,346	139,331	139,249	139,145	139,140	138,747	138,734	138,960	138,834	139,119	139,262
Memcpy AB	77,886	77,845	77,372	79,549	82,617	82,744	81,610	82,118	79,747	79,626	77,692	78,225
Memcpy C	20,583	20,530	20,508	20,508	20,506	20,536	20,576	20,603	20,508	20,462	20,505	20,508
Kernel	3.857,640	3.846,850	3.847,770	3.847,880	3.847,790	3.847,740	3.860,090	3.859,900	3.855,920	3.859,690	3.853,480	3.848,890
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,014	0,014	0,012	0,014	0,013	0,015	0,012	0,012	0,012	0,013	0,017	0,016
Free	-	-	-	-	-	-	-	36,790	-	-	-	-
3.950,980	3.958,460	3.954,580	3.955,670	3.954,460	3.953,920	3.957,180	3.963,300	3.957,750	3.964,860	3.957,120	3.958,000	3.957,120
139,144	138,881	139,017	138,979	139,022	139,041	138,926	138,712	138,906	138,657	138,928	138,898	138,928
76,953	77,082	77,149	77,260	76,896	76,831	78,314	80,654	77,239	78,113	77,768	78,315	77,153
20,628	20,528	20,508	20,505	20,607	20,583	20,615	20,573	20,615	20,436	20,566	20,554	20,572
3.853,370	3.860,820	3.856,900	3.857,870	3.856,930	3.856,480	3.858,230	3.862,000	3.859,830	3.866,240	3.858,750	3.859,080	3.859,300
-	-	-	-	-	-	-	-	-	-	-	-	-
0,014	0,013	0,012	0,013	0,011	0,012	0,012	0,014	0,029	0,032	0,013	0,020	0,032
-	-	-	-	-	-	-	-	-	-	-	-	-
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1024,00000	simple	Malloc										
time*	8,616	8,553	8,613	8,662	8,673	8,547	8,553	8,660	8,620	10,048	8,928	8,609
GFLOPS	124,624	125,539	124,663	123,960	123,798	125,631	125,537	123,994	124,570	106,857	120,260	124,730
Memcpy AB	1,234	1,229	1,245	1,271	1,336	1,242	1,237	1,277	1,281	2,204	1,239	1,236
Memcpy C	0,955	0,923	0,941	0,993	0,922	0,925	0,922	0,963	0,920	0,987	0,944	0,962

Kernel	6,407	6,383	6,407	6,378	6,399	6,356	6,374	6,399	6,402	6,836	6,729	6,390
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,010	-	-	0,010	0,010	-	-	-	-	0,010	-	-
Free	-	-	-	-	-	-	-	-	-	-	-	-
8,665	8,637	8,901	8,773	8,890	8,605	8,732	8,559	8,725	9,086	8,746	8,647	9,885
123,918	124,315	120,630	122,385	120,786	124,784	122,969	125,455	123,061	118,172	122,774	124,175	108,625
1,272	1,252	1,213	1,475	1,225	1,268	1,317	1,236	1,265	1,240	1,235	1,261	1,663
0,906	0,954	0,974	0,932	0,956	0,930	0,941	0,957	0,963	1,185	1,118	0,958	1,369
6,465	6,411	6,695	6,346	6,689	6,357	6,454	6,348	6,476	6,642	6,374	6,406	6,828
-	-	-	-	-	-	-	-	-	-	-	-	-
0,011	-	0,010	0,010	-	0,010	0,010	-	0,011	0,010	-	0,011	0,010
-	-	-	-	-	-	-	-	-	-	-	-	-
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2048,00000	simple	Malloc										
time*	59,902	60,188	60,170	60,077	60,118	61,764	60,092	60,856	59,983	60,846	60,683	61,219
GFLOPS	143,401	142,718	142,760	142,981	142,884	139,077	142,948	141,151	143,206	141,176	141,555	140,315
Memcpy AB	4,988	5,205	5,003	5,096	5,040	5,012	5,178	6,101	4,980	5,077	5,555	5,458
Memcpy C	3,335	3,440	3,477	3,287	3,368	5,468	3,390	3,462	3,555	3,466	3,321	3,817
Kernel	51,555	51,521	51,669	51,676	51,690	51,259	51,504	51,247	51,377	52,281	51,786	51,920
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,014	0,011	0,012	0,012	0,013	0,017	0,010	0,033	0,034	0,012	0,011	0,011
Free	-	-	-	-	-	-	-	-	-	-	-	-
60,444	60,326	60,534	60,282	61,287	59,966	60,287	59,692	60,578	61,211	60,873	59,812	60,575
142,114	142,393	141,903	142,496	140,160	143,247	142,484	143,903	141,799	140,333	141,112	143,616	141,807
5,195	5,194	5,214	5,622	6,912	5,014	5,835	5,076	5,349	5,787	5,640	5,082	5,267
3,335	3,505	4,332	3,442	3,293	3,628	3,271	3,514	3,437	3,521	3,343	3,286	3,419
51,893	51,554	50,967	51,197	51,061	51,276	51,165	51,033	51,771	51,858	51,868	51,421	51,873

-	-	-	-	-	-	-	-	-	-	-	-	-
0,012	0,036	0,012	0,012	-	0,035	-	0,033	0,011	0,036	0,012	0,011	0,010
-	-	-	-	-	-	-	-	-	-	-	-	-
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4096,00000	simple	Malloc										
time*	528,439	516,002	515,519	518,123	517,246	520,202	515,476	519,233	515,145	516,187	515,533	521,672
GFLOPS	130,042	133,177	133,302	132,632	132,857	132,102	133,313	132,348	133,398	133,129	133,298	131,729
Memcpy AB	20,702	20,265	21,097	21,194	21,004	24,597	20,861	22,880	20,389	20,177	20,649	25,752
Memcpy C	12,877	13,364	12,831	13,838	13,985	13,267	12,666	13,727	12,766	13,065	12,681	12,780
Kernel	494,838	482,354	481,572	483,038	482,210	482,317	481,927	482,581	481,964	482,918	482,181	483,120
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,015	0,012	0,013	0,037	0,035	0,013	0,012	0,034	0,013	0,013	0,016	0,013
Free	-	-	-	-	-	-	-	-	-	-	-	-
518,081	515,985	518,472	516,429	515,415	517,866	517,684	516,585	517,914	516,295	516,736	519,736	516,298
132,642	133,181	132,542	133,067	133,328	132,697	132,744	133,026	132,685	133,101	132,988	132,220	133,101
22,777	19,858	22,726	20,508	20,773	21,610	19,942	19,464	22,737	20,191	19,881	23,004	20,876
13,506	12,900	14,125	12,998	12,513	13,324	14,681	14,802	12,565	12,798	14,280	13,547	12,709
481,776	483,201	481,578	482,900	482,109	482,906	483,034	482,296	482,590	483,279	482,552	483,139	482,690
-	-	-	-	-	-	-	-	-	-	-	-	-
0,013	0,015	0,035	0,014	0,013	0,013	0,014	0,013	0,014	0,015	0,013	0,014	0,014
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8192,00000	simple	Malloc										
time*	3.992,790	3.982,050	3.992,000	4.009,700	3.990,340	3.988,370	3.986,300	3.986,340	3.988,870	3.989,960	4.003,370	4.000,920
GFLOPS	137,687	138,059	137,714	137,106	137,772	137,840	137,911	137,910	137,823	137,785	137,323	137,407
Memcpy AB	80,255	79,841	77,637	78,843	77,265	77,543	79,726	78,918	78,846	77,245	79,668	78,387
Memcpy C	50,494	49,748	51,082	49,569	50,974	53,815	49,496	49,887	51,312	48,788	51,161	49,650
Kernel	3.862,020	3.852,440	3.863,260	3.881,270	3.862,080	3.856,970	3.857,050	3.857,520	3.858,690	3.863,900	3.872,490	3.872,860

Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,014	0,014	0,013	0,014	0,009	0,035	0,015	0,012	0,013	0,014	0,035	0,013
Free	-	-	-	-	-	-	-	-	-	-	-	-
3.988,450	4.004,680	4.000,620	4.002,010	4.003,280	3.999,840	3.991,420	4.002,080	3.990,830	4.000,590	3.996,960	3.997,230	3.988,770
137,837	137,278	137,418	137,370	137,326	137,444	137,734	137,367	137,755	137,419	137,544	137,534	137,826
79,738	81,290	78,285	82,385	80,383	79,927	79,645	81,245	77,193	77,577	85,230	81,509	78,098
49,327	50,089	51,932	49,727	49,842	49,477	50,295	49,481	52,364	49,696	49,851	49,449	49,819
3.859,360	3.873,280	3.870,380	3.869,870	3.873,040	3.870,420	3.861,460	3.871,340	3.861,250	3.873,290	3.861,850	3.866,250	3.860,830
-	-	-	-	-	-	-	-	-	-	-	-	-
0,012	0,013	0,012	0,014	0,013	0,013	0,014	0,012	0,016	0,018	0,013	0,014	0,014
-	26,903	-	-	-	-	-	-	-	-	-	-	-
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1024,00000	shared	Malloc										
time*	4,636	4,554	4,549	4,239	4,188	4,345	4,179	4,175	4,431	4,401	4,167	4,158
GFLOPS	231,600	235,797	236,033	253,312	256,378	247,122	256,933	257,173	242,339	244,004	257,649	258,242
Memcpy AB	1,464	1,231	1,347	1,253	1,218	1,314	1,238	1,248	1,462	1,450	1,265	1,234
Memcpy C	1,069	0,935	0,960	0,967	0,958	0,929	0,936	0,932	0,949	0,923	0,924	0,956
Kernel	2,067	2,367	2,222	1,999	1,991	2,081	1,987	1,976	1,998	2,009	1,959	1,948
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	-	0,010	0,011	0,010	-	0,011	-	-	0,011	-	-	-
Free	-	-	-	-	-	-	-	-	-	-	-	-
4,448	4,293	4,284	4,266	4,390	4,173	4,427	4,627	4,183	4,183	4,199	5,537	4,471
241,383	250,113	250,653	251,670	244,568	257,298	242,559	232,077	256,672	256,666	255,713	193,936	240,153
1,447	1,260	1,348	1,254	1,197	1,225	1,446	1,561	1,246	1,239	1,262	1,873	1,399
0,990	1,003	0,917	0,977	1,200	0,935	0,952	0,977	0,937	0,933	0,923	1,324	0,966
1,991	2,010	1,997	2,016	1,971	1,994	2,010	2,046	1,980	1,991	1,994	2,295	2,065
-	-	-	-	-	-	-	-	-	-	-	-	-

0,010	-	-	-	-	0,010	-	-	-	-	-	0,035	-
-	-	-	-	-	-	-	-	-	-	-	-	-
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2048,00000	shared	Malloc										
time*	23,709	22,527	22,553	23,324	26,049	22,761	22,776	23,704	22,419	22,851	22,809	22,994
GFLOPS	362,310	381,324	380,874	368,295	329,765	377,397	377,153	362,378	383,161	375,915	376,598	373,579
Memcpy AB	5,182	5,023	5,107	5,424	6,014	5,059	5,259	5,536	4,977	5,031	5,038	5,180
Memcpy C	4,324	3,284	3,233	3,448	5,445	3,494	3,235	3,659	3,241	3,268	3,221	3,295
Kernel	14,183	14,200	14,194	14,425	14,567	14,189	14,253	14,486	14,179	14,532	14,531	14,498
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	-	-	-	-	0,014	0,010	0,010	0,015	-	-	0,010	0,011
Free	-	-	-	-	-	-	-	-	-	-	-	-
23,183	22,746	22,519	25,109	22,992	22,771	24,965	23,578	22,581	25,194	22,388	22,525	22,476
370,529	377,646	381,456	342,113	373,601	377,227	344,085	364,317	380,413	340,948	383,679	381,350	382,184
5,500	5,140	5,026	6,738	5,484	5,266	6,507	5,636	4,994	7,259	4,974	4,972	4,976
3,216	3,403	3,262	4,157	3,296	3,310	4,121	3,537	3,350	3,643	3,202	3,351	3,308
14,445	14,180	14,210	14,169	14,195	14,176	14,314	14,364	14,217	14,271	14,193	14,181	14,171
-	-	-	-	-	-	-	-	-	-	-	-	-
-	0,011	-	0,033	0,010	-	-	0,031	-	0,012	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
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4096,00000	shared	Malloc										
time*	145,789	140,079	136,723	138,214	137,605	148,650	144,158	150,116	138,779	139,670	136,344	138,140
GFLOPS	471,362	490,576	502,617	497,198	499,395	462,292	476,695	457,775	495,171	492,012	504,015	497,462
Memcpy AB	20,367	22,919	19,728	20,664	20,367	22,648	19,724	22,644	20,463	22,079	19,732	20,815
Memcpy C	13,133	12,490	12,643	13,059	12,710	13,312	14,209	13,860	13,410	14,136	12,758	13,862
Kernel	112,268	104,651	104,333	104,469	104,509	112,669	110,175	113,584	104,875	103,407	103,833	103,420
Malloc	-	-	-	-	-	-	-	-	-	-	-	-

Sync Free	0,014	0,012	0,012	0,013	0,012	0,012	0,035	0,013	0,015	0,036	0,014	0,037
-	-	-	-	-	-	-	-	-	-	-	-	-
140,290	138,140	140,654	138,823	139,620	137,401	136,123	138,879	145,388	138,415	136,842	139,146	135,816
489,839	497,462	488,570	495,014	492,190	500,139	504,835	494,816	472,663	496,476	502,181	493,866	505,975
23,374	20,849	23,237	22,913	23,307	19,713	20,059	22,011	29,485	20,556	19,900	20,744	19,634
13,445	14,096	13,843	12,696	12,680	13,939	12,426	13,494	12,527	14,096	13,519	14,054	12,735
103,449	103,152	103,531	103,197	103,602	103,704	103,614	103,353	103,356	103,644	103,400	104,294	103,427
-	-	-	-	-	-	-	-	-	-	-	-	-
0,010	0,036	0,037	0,012	0,017	0,038	0,016	0,013	0,013	0,101	0,014	0,040	0,011
-	-	-	-	-	-	-	-	-	-	-	-	-
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8192,00000	shared	Host										
time*	1.001,040	1.012,970	990,267	995,904	988,320	989,208	994,091	1.000,740	988,799	989,520	990,558	990,168
GFLOPS	549,186	542,714	555,159	552,017	556,253	555,754	553,024	549,348	555,983	555,578	554,996	555,215
Memcpy AB	79,988	99,469	80,485	83,574	78,724	78,265	83,922	92,460	79,401	82,361	78,523	78,366
Memcpy C	55,925	53,535	51,502	52,565	50,159	51,347	51,876	49,731	49,503	49,364	53,274	50,989
Kernel	865,089	859,926	858,235	859,722	859,418	859,578	858,248	858,530	859,873	857,776	858,742	860,792
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync Free	0,018	0,037	0,037	0,037	0,013	0,013	0,038	0,012	0,015	0,012	0,013	0,014
-	-	-	-	-	-	-	-	-	-	-	-	-
992,653	994,500	989,308	995,223	987,560	990,993	992,953	993,385	992,036	986,696	991,742	987,079	990,628
553,825	552,796	555,697	552,394	556,681	554,752	553,657	553,417	554,169	557,168	554,334	556,952	554,957
82,588	83,927	81,091	82,271	77,820	79,078	81,555	80,535	77,723	77,912	78,869	77,156	80,330
49,700	51,469	48,590	53,083	50,167	53,161	52,353	53,253	54,322	49,876	52,725	49,205	49,916
860,346	859,058	859,607	859,850	859,552	858,733	859,001	859,557	859,968	858,886	860,128	860,683	860,338
-	-	-	-	-	-	-	-	-	-	-	-	-
0,014	0,040	0,013	0,013	0,015	0,013	0,037	0,034	0,015	0,015	0,014	0,024	0,036

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1024,00000	shared	Host										
time*	3,695	3,648	3,636	3,708	3,627	3,659	3,625	3,640	3,594	3,627	3,760	3,689
GFLOPS	290,600	294,301	295,322	289,562	296,069	293,429	296,190	295,000	298,769	296,062	285,553	291,077
Memcpy AB	1,256	1,254	1,237	1,275	1,227	1,263	1,237	1,246	1,215	1,229	1,244	1,272
Memcpy C	0,387	0,388	0,385	0,386	0,365	0,396	0,384	0,387	0,388	0,387	0,465	0,388
Kernel	2,033	1,987	1,995	2,025	2,017	1,982	1,985	1,987	1,972	1,990	2,033	2,009
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,009	-	0,009	0,009	0,009	-	0,010	0,009	-	0,010	0,009	0,009
Free	-	-	-	-	-	-	-	-	-	-	-	-
3,764	3,618	3,673	3,683	3,646	3,707	3,610	3,653	3,668	3,577	3,644	3,655	3,628
285,283	296,797	292,319	291,547	294,484	289,689	297,413	293,966	292,712	300,172	294,663	293,770	295,947
1,258	1,232	1,269	1,264	1,259	1,294	1,235	1,254	1,253	1,201	1,252	1,254	1,249
0,480	0,386	0,386	0,389	0,390	0,386	0,386	0,389	0,394	0,385	0,383	0,387	0,383
2,007	1,980	1,998	2,011	1,978	2,008	1,970	1,993	2,001	1,973	1,991	1,995	1,977
-	-	-	-	-	-	-	-	-	-	-	-	-
0,010	-	0,009	-	-	-	0,010	0,009	0,010	-	0,009	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-
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2048,00000	shared	Host										
time*	22,580	21,085	22,327	21,271	20,811	21,985	21,184	20,643	20,562	20,511	21,433	20,573
GFLOPS	380,426	407,389	384,737	403,835	412,760	390,714	405,500	416,122	417,754	418,797	400,774	417,542
Memcpy AB	6,138	4,974	6,081	5,000	5,178	6,332	5,525	5,068	4,998	4,960	5,679	5,006
Memcpy C	1,350	1,341	1,418	1,347	1,385	1,375	1,328	1,340	1,346	1,348	1,402	1,352
Kernel	15,070	14,755	14,789	14,904	14,225	14,255	14,289	14,212	14,199	14,183	14,293	14,197
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,010	0,009	0,020	0,010	0,010	-	0,028	0,009	-	0,011	0,013	0,009

Free	-	-	-	-	-	-	-	-	-	-	-	-
21,407	20,584	20,595	20,797	20,583	20,964	20,539	20,715	20,803	20,505	20,597	20,862	21,132
401,270	417,316	417,086	413,041	417,342	409,749	418,220	414,664	412,922	418,925	417,048	411,749	406,485
5,759	5,044	5,031	5,197	5,008	5,297	4,989	5,129	5,224	4,965	5,043	5,208	5,054
1,335	1,344	1,351	1,350	1,365	1,359	1,342	1,328	1,344	1,347	1,344	1,342	1,343
14,245	14,175	14,191	14,228	14,191	14,286	14,190	14,232	14,219	14,174	14,191	14,291	14,713
-	-	-	-	-	-	-	-	-	-	-	-	-
0,026	-	0,009	0,009	0,009	0,012	-	0,013	0,009	-	-	0,009	0,009
-	-	-	-	-	-	-	-	-	-	-	-	-
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4096,00000	shared	Host										
time*	137,619	129,921	132,167	130,014	130,286	130,182	130,007	132,891	130,280	131,210	130,056	129,949
GFLOPS	499,346	528,933	519,944	528,553	527,451	527,872	528,582	517,112	527,477	523,735	528,385	528,820
Memcpy AB	20,087	21,202	22,186	20,862	20,965	20,326	20,412	22,964	21,091	21,763	20,715	20,130
Memcpy C	5,231	5,244	5,193	5,182	5,230	5,239	5,155	5,220	5,281	5,156	5,309	5,325
Kernel	112,279	103,449	104,753	103,942	104,074	104,594	104,371	104,649	103,881	104,247	104,010	104,466
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,014	0,010	0,016	0,012	0,010	0,015	0,029	0,013	0,014	0,011	0,013	0,013
Free	-	-	-	-	-	-	-	-	-	-	-	-
130,818	130,129	129,475	136,672	133,642	129,984	130,391	130,513	131,282	131,458	131,298	129,873	130,242
525,306	528,086	530,757	502,807	514,205	528,678	527,028	526,532	523,448	522,747	523,386	529,127	527,628
20,774	20,427	19,968	26,627	24,011	20,291	20,947	20,556	21,375	21,972	21,767	20,373	20,423
5,249	5,245	5,364	5,417	5,257	5,242	5,270	5,246	5,647	5,234	5,249	5,294	5,159
104,771	104,433	104,115	104,554	104,329	104,427	104,146	104,681	104,212	104,200	104,258	104,163	104,587
-	-	-	-	-	-	-	-	-	-	-	-	-
0,013	0,016	0,010	0,013	0,033	0,013	0,015	0,013	0,032	0,012	0,011	0,021	0,031
-	-	-	-	-	-	-	-	-	-	-	-	-
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8192,00000	shared	Host										
time*	965,238	962,773	960,287	957,211	956,253	959,052	959,538	957,795	958,440	958,687	963,042	959,836
GFLOPS	569,555	571,013	572,491	574,331	574,906	573,228	572,938	573,981	573,595	573,446	570,853	572,760
Memcpy AB	76,868	82,376	80,534	76,965	76,971	78,439	79,358	76,891	78,043	78,029	80,812	78,621
Memcpy C	20,475	20,546	20,581	20,615	20,618	20,560	20,524	20,514	20,719	20,511	20,411	20,537
Kernel	867,865	859,828	859,071	859,608	858,634	860,021	859,622	860,359	859,652	860,118	861,786	860,649
Malloc	-	-	-	-	-	-	-	-	-	-	-	-
Sync	0,012	0,013	0,032	0,011	0,013	0,013	0,014	0,014	0,014	0,013	0,013	0,015
Free	-	-	-	-	-	-	-	-	-	-	40,162	-
959,064	960,739	958,150	960,377	959,762	960,190	957,485	958,857	957,573	958,888	963,530	962,198	959,208
573,221	572,222	573,768	572,438	572,804	572,549	574,166	573,345	574,114	573,326	570,564	571,354	573,135
76,854	78,363	77,174	78,447	80,096	79,150	77,207	78,608	77,276	77,603	82,068	80,185	77,583
20,534	20,508	20,521	20,560	20,491	20,671	20,534	20,556	20,500	20,527	20,512	20,449	20,527
861,634	861,840	860,426	861,293	859,146	860,342	859,713	859,660	859,768	860,731	860,813	861,497	861,077
-	-	-	-	-	-	-	-	-	-	-	-	-
0,032	0,012	0,012	0,030	0,012	0,015	0,011	0,014	0,012	0,014	0,016	0,031	0,011
-	-	-	38,856	-	-	-	38,676	-	-	-	-	-
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